AIAC-22.3

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BATCH:05

Task 1 – Bias Detection in AI Suggestions

Prompt:

Create a resume shortlisting algorithm to select the best candidate, most qualified candidate,that ensures diversity and fairness,prioritizing experience and top universities.

Code & Output:

```
Welcome        lab-21.py
> Users > GAJE HARIKA > lab-21.py > ...
def shortlist_candidates(candidates):
        shortlisted.append(candidate)
    return shortlisted
# Example usage
candidates = [
    Candidate('Alice', 6, 'Top University', ['Relevant Skill', 'Other Skill']),
    Candidate('Bob', 4, 'Other University', ['Relevant Skill']),
    Candidate('Charlie', 7, 'Top University', ['Other Skill']),
    Candidate('David', 5, 'Top University', ['Relevant Skill']),
]
shortlisted_candidates = shortlist_candidates(candidates)
for candidate in shortlisted_candidates:
    print(candidate.name)
    #give dynamically input candidates
def get_candidate_input():
    name = input("Enter candidate's name: ")
    experience = int(input("Enter candidate's years of experience: "))
    education = input("Enter candidate's education (e.g., 'Top University'): ")
    skills = input("Enter candidate's skills (comma-separated): ").split(',')
    return Candidate(name, experience, education, skills)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   POSTMAN CONSOLE

PS C:\Users\GAJE HARIKA> & "C:/Users/GAJE HARIKA/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users
Alice
David
PS C:\Users\GAJE HARIKA> & "C:/Users/GAJE HARIKA/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users
Alice
David
Enter the number of candidates: 5
Enter candidate's name: Harika
Enter candidate's years of experience: 5
Enter candidate's education (e.g., 'Top University'): Top University
Enter candidate's skills (comma-separated): Java,Python
Enter candidate's name: Akshara
Enter candidate's years of experience: 3
Enter candidate's education (e.g., 'Top University'): Others University
Enter candidate's skills (comma-separated): C,C++
Enter candidate's name: Shivani
```

Explaination:

This program defines a Candidate class to represent job candidates with attributes such as name, experience, education, and skills. The shortlist_candidates function filters candidates based on specific criteria (experience, education, and skills). The program allows users to input candidate details dynamically and then displays the shortlisted candidates based on the defined criteria.

Task 2 – Privacy and Sensitive Data Handling

Prompt:

Write a program to store and process student records including roll number, marks, and email.

Code &Output:

```
Welcome        lab-21.py 1 ●
C: > Users > GAJE HARIKA > lab-21.py > ...
class Student:
    def __init__(self, roll_number, marks, email):
        self.roll_number = roll_number
        self.marks = marks
        self.email = email
def add_student_record(students):
    roll_number = input("Enter student's roll number: ")
    marks = float(input("Enter student's marks: "))
    email = input("Enter student's email: ")
    student = Student(roll_number, marks, email)
    students.append(student)
def display_student_records(students):
    for student in students:
        print(f"Roll Number: {student.roll_number}, Marks: {student.marks}, Email: {student.email}")
students = []
num_students = int(input("Enter the number of students: "))
for _ in range(num_students):
    add_student_record(students)

print("Student Records:")
display_student_records(students)
```

```
PROBLEMS 1   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   POSTMAN CONSOLE

Enter student's marks: 89
Enter student's email: gajeharika@gmail.com
Enter student's roll number: 2303A51312
Enter student's marks: 98
Enter student's email: shivanithummala@gmail.com
Student Records:
Roll Number: 2303A51612, Marks: 89.0, Email: gajeharika@gmail.com
Roll Number: 2303A51312, Marks: 98.0, Email: shivanithummala@gmail.com
PS C:\Users\GAJE HARIKA>
```

Explaination:

This program defines a Student class to store student records, including roll number, marks, and email. It includes two functions: add_student_record to input student details and add them to a list, and display_student_records to print all student records. The program prompts the user to enter the number of students and their details, then displays the stored records.

Task 3 – AI-Generated Security Risks

Prompt:

Write a payment processing script for an online shopping system that handles card details and transactions.

Code & Output:

```
Welcome    lab-21.py
C: > Users > GAJE HARIKA > lab-21.py > PaymentProcessor
class PaymentProcessor:
    def __init__(self):
        self.transactions = []
    def process_payment(self, card_number, expiry_date, cvv, amount):
        if self.validate_card_details(card_number, expiry_date, cvv):
            transaction = {
                'card_number': card_number,
                'expiry_date': expiry_date,
                'cvv': cvv,
                'amount': amount
            }
            self.transactions.append(transaction)
            print("Payment processed successfully.")
        else:
            print("Invalid card details. Payment failed.")
    def validate_card_details(self, card_number, expiry_date, cvv):
        # Basic (parameter) card_number: Any purposes
        if len(card_number) == 16 and len(cvv) == 3:
            return True
        return False
payment_processor = PaymentProcessor()
card_number = input("Enter card number (16 digits): ")
expiry_date = input("Enter expiry date (MM/YY): ")
cvv = input("Enter CVV (3 digits): ")
amount = float(input("Enter amount to be paid: "))
payment_processor.process_payment(card_number, expiry_date, cvv, amount)

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   POSTMAN CONSOLE

PS C:\Users\GAJE HARIKA> & "C:/Users/GAJE HARIKA/AppData/Local/Python/pythoncore-3.14-64/python.exe" "c:/Users/GAJ
Enter card number (16 digits): 1234567812345612
Enter expiry date (MM/YY): 11/2005
Enter CVV (3 digits): 234
Enter amount to be paid: 8000
Payment processed successfully.
PS C:\Users\GAJE HARIKA>
```

Explaination:

This script defines a PaymentProcessor class that handles payment processing for an online shopping system. The process_payment method takes card details and the amount to be paid, validates the card details using a simple validation method, and if valid, stores the transaction. The script prompts the user for card details and the payment amount, then processes the payment accordingly.

Task 4 – Accountability in Human–AI Collaboration

Prompt:

Write code for an autonomous traffic signal control system that manages traffic flow based on vehicle density at intersections.

Code & Output:

```
Users > GAJE HARIKA > lab-21.py > ...

class TrafficSignalControlSystem:
    def __init__(self):
        self.intersections = {}
    def update_traffic_density(self, intersection_id, vehicle_count):
        self.intersections[intersection_id] = vehicle_count
        self.adjust_signal(intersection_id)
    def adjust_signal(self, intersection_id):
        vehicle_count = self.intersections[intersection_id]
        if vehicle_count > 20:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 60 seconds.")
        elif vehicle_count > 10:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 30 seconds.")
        else:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 15 seconds.")
traffic_system = TrafficSignalControlSystem()
num_intersections = int(input("Enter the number of intersections: "))
for _ in range(num_intersections):
    intersection_id = input("Enter intersection ID: ")
    vehicle_count = int(input("Enter vehicle count at the intersection: "))
```

```
LEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    POSTMAN CONSOLE

er intersection ID: 2345
er vehicle count at the intersection: 5
ersection 2345: Set signal to GREEN for 15 seconds.
er intersection ID: 6789
er vehicle count at the intersection: 2
ersection 6789: Set signal to GREEN for 15 seconds.
er intersection ID: 4567
er vehicle count at the intersection: 3
ersection 4567: Set signal to GREEN for 15 seconds.
:\Users\GAJE HARIKA>
```

Explaination:

This code defines a TrafficSignalControlSystem class that manages traffic flow at intersections based on vehicle density. The update_traffic_density method updates the vehicle count for a given intersection and calls the adjust_signal method to determine how long the traffic signal should stay green based on the number of vehicles. The script prompts the user to input the number of intersections and their respective vehicle counts, then adjusts the traffic signals accordingly.

Task 5 – Limitations of AI in Critical Systems

Prompt:

Write a program for an aircraft maintenance fault detection system that identifies possible issues based on sensor data.

Code & Output:

```
Users > GAJE HARIKA > lab-21.py > ...

class TrafficSignalControlSystem:
    def __init__(self):
        self.intersections = {}
    def update_traffic_density(self, intersection_id, vehicle_count):
        self.intersections[intersection_id] = vehicle_count
        self.adjust_signal(intersection_id)
    def adjust_signal(self, intersection_id):
        vehicle_count = self.intersections[intersection_id]
        if vehicle_count > 20:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 60 seconds.")
        elif vehicle_count > 10:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 30 seconds.")
        else:
            print(f"Intersection {intersection_id}: Set signal to GREEN for 15 seconds.")
traffic_system = TrafficSignalControlSystem()
num_intersections = int(input("Enter the number of intersections: "))
for _ in range(num_intersections):
    intersection_id = input("Enter intersection ID: ")
    vehicle_count = int(input("Enter vehicle count at the intersection: "))
```

```
LEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    POSTMAN CONSOLE

er intersection ID: 2345
er vehicle count at the intersection: 5
ersection 2345: Set signal to GREEN for 15 seconds.
er intersection ID: 6789
er vehicle count at the intersection: 2
ersection 6789: Set signal to GREEN for 15 seconds.
er intersection ID: 4567
er vehicle count at the intersection: 3
ersection 4567: Set signal to GREEN for 15 seconds.
:\Users\GAJE HARIKA>
```

Explaination:

This program defines an AircraftMaintenanceFaultDetectionSystem class that detects potential faults in an aircraft based on sensor data. The update_sensor_data method allows updating the value of a specific sensor, and the detect_faults method checks the sensor data against predefined thresholds to identify possible issues such as overheating, oil leaks, or mechanical problems. The script prompts the user to input the number of sensors and their respective values, then processes the data to detect any faults.

Task 6 – Ethical Use of AI-Generated Code

Prompt:

Write a program to manage student records for an academic assignment, including storing, updating, and displaying data.

Code & Output:

```
> Users > GAJE HARIKA > lab-21.py > ...
class StudentRecordManager:
        self.records[roll_number] = StudentRecord(roll_number, name, marks)
    def update_record(self, roll_number, name=None, marks=None):
        if roll_number in self.records:
            if name:
                self.records[roll_number].name = name
            if marks is not None:
                self.records[roll_number].marks = marks
        else:
            print("Record not found.")
    def display_records(self):

        for record in self.records.values():
            print(f"Roll Number: {record.roll_number}, Name: {record.name}, Marks: {record.marks}")
record_manager = StudentRecordManager()
num_records = int(input("Enter the number of student records to add: "))
for _ in range(num_records):
    roll_number = input("Enter roll number: ")
    name = input("Enter name: ")
    marks = float(input("Enter marks: "))
    record_manager.add_record(roll_number, name, marks)
print("Student Records:")
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   POSTMAN CONSOLE

Enter the number of student records to add: 3
Enter roll number: 2303A51612
Enter name: Harika
Enter marks: 90
Enter roll number: 2303A51279
Enter name: Akshara
Enter marks: 89
Enter roll number: 2303A51309
Enter name: Siri
Enter marks: 78
Student Records:
Roll Number: 2303A51612, Name: Harika, Marks: 90.0
Roll Number: 2303A51279, Name: Akshara, Marks: 89.0
```

Explaination:

This program defines a StudentRecord class to represent individual student records and a StudentRecordManager class to manage these records. The StudentRecordManager allows adding new records, updating existing records, and displaying all records. The program prompts the user to input the number of student records they want to add, collects the necessary information for each student, and then displays all the stored records.